2.31 Integration: Area Between Curves

Worked Solutions 1–12

1)

$$x^2 = x^3 \Rightarrow x^3 - x^2 = 0 \Rightarrow x^2(x-1) = 0 \Rightarrow x = 0 \text{ or } x = 1$$

$$\begin{aligned}\int_0^1 x^2\,dx - \int_0^1 x^3\,dx = \int_0^1 (x^2 - x^3)\,dx &= [\tfrac{1}{3}x^3 - \tfrac{1}{4}x^4]_0^1 \\ &= (\tfrac{1}{3}\times 1^3 - \tfrac{1}{4}\times 1^4) - (\tfrac{1}{3}\times 0^3 - \tfrac{1}{4}\times 0^4) \\ &= (\tfrac{1}{12}) - (0) = \tfrac{1}{12}\end{aligned}$$

2)

$$4 - x^2 = x^2 + 2 \Rightarrow 2x^2 = 2 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\begin{aligned}\int_{-1}^1 (4 - x^2)\,dx - \int_{-1}^1 (x^2 + 2)\,dx = \int_{-1}^1 (2 - 2x^2)\,dx &= [2x - \tfrac{2}{3}x^3]_{-1}^1 \\ &= (2\times 1 - \tfrac{2}{3}\times 1^3) - (2\times(-1) - \tfrac{2}{3}\times(-1)^3) \\ &= (\tfrac{4}{3}) - (-\tfrac{4}{3}) = \tfrac{8}{3}\end{aligned}$$

3)

$$\begin{aligned}5 - x^2 = (x-1)^2 &\Rightarrow 5 - x^2 = x^2 - 2x + 1 \\ &\Rightarrow 2x^2 - 2x - 4 = 0 \Rightarrow x^2 - x - 2 = 0 \\ &(x-2)(x+1) = 0 \Rightarrow x = -1 \text{ or } x = 2\end{aligned}$$

$$\begin{aligned}\int_{-1}^2 (5 - x^2)\,dx - \int_{-1}^2 (x-1)^2\,dx &= \int_{-1}^2 (5 - x^2 - (x-1)^2)\,dx \\ &= \int_{-1}^2 (5 - x^2 - (x^2 - 2x + 1))\,dx \\ &= \int_{-1}^2 (-2x^2 + 2x + 4)\,dx = [-\tfrac{2}{3}x^3 + x^2 + 4x]_{-1}^2 \\ &= (-\tfrac{2}{3}\times 2^3 + 2^2 + 4\times 2) \\ &\quad - (-\tfrac{2}{3}\times(-1)^3 + (-1)^2 + 4\times(-1)) \\ &= (\tfrac{20}{3}) - (-\tfrac{7}{3}) = 9\end{aligned}$$

4)

$$\begin{aligned}(x-1)^2 + 2 &= 6 - (x+1)^2 \\ \Rightarrow (x^2 - 2x + 1) + 2 &= 6 - (x^2 + 2x + 1) \\ \Rightarrow x^2 - 2x + 3 &= 5 - x^2 - 2x \Rightarrow 2x^2 = 2\end{aligned}$$

$$\Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\begin{aligned}\int_{-1}^{1} (6-(x+1)^2)\,dx - \int_{-1}^{1} ((x-1)^2+2)\,dx & \\ &= \int_{-1}^{1} (4-(x+1)^2-(x-1)^2)\,dx \\ &= \int_{-1}^{1} (4-(x^2+2x+1)-(x^2-2x+1))\,dx \\ &= \int_{-1}^{1} (2-2x^2)\,dx = [2x-\tfrac{2}{3}x^3]_{-1}^{1} \\ &= (2\times 1-\tfrac{2}{3}\times 1^3)-(2\times(-1)-\tfrac{2}{3}\times(-1)^3) \\ &= (\tfrac{4}{3})-(-\tfrac{4}{3}) = \tfrac{8}{3}\end{aligned}$$

5)

$$x^3+1 = 3-x^3 \Rightarrow 2x^3 = 2 \Rightarrow x^3 = 1 \Rightarrow x = 1$$

$$\begin{aligned}\int_0^1 (3-x^3)\,dx - \int_0^1 (x^3+1)\,dx = \int_0^1 (2-2x^3)\,dx &= [2x-\tfrac{1}{2}x^4]_0^1 \\ &= (2\times 1-\tfrac{1}{2}\times 1^4)-(2\times 0-\tfrac{1}{2}\times 0^4) \\ &= (\tfrac{3}{2})-(0) = \tfrac{3}{2}\end{aligned}$$

6)

$$x^2 = \sqrt{x} \Rightarrow x^4 = x \Rightarrow x^4 - x = 0 \Rightarrow x(x^3-1) = 0 \Rightarrow x = 0 \text{ or } x = 1$$

$$\begin{aligned}\int_0^1 \sqrt{x}\,dx - \int_0^1 x^2\,dx = \int_0^1 (x^{1/2}-x^2)\,dx &= [\tfrac{2}{3}x^{3/2}-\tfrac{1}{3}x^3]_0^1 \\ &= (\tfrac{2}{3}\times 1^{3/2}-\tfrac{1}{3}\times 1^3)-(\tfrac{2}{3}\times 0^{3/2}-\tfrac{1}{3}\times 0^3) \\ &= (\tfrac{1}{3})-(0) = \tfrac{1}{3}\end{aligned}$$

7)

$$1-x^2 = \tfrac{4}{3}(1-x^2) \Rightarrow 3-3x^2 = 4-4x^2 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\begin{aligned}\int_{-1}^{1} \tfrac{4}{3}(1-x^2)\,dx - \int_{-1}^{1} (1-x^2)\,dx = \int_{-1}^{1} \tfrac{1}{3}(1-x^2)\,dx & \\ &= \int_{-1}^{1} (\tfrac{1}{3}-\tfrac{1}{3}x^2)\,dx = [\tfrac{1}{3}x-\tfrac{1}{9}x^3]_{-1}^{1} \\ &= (\tfrac{1}{3}\times 1-\tfrac{1}{9}\times 1^3)-(\tfrac{1}{3}\times(-1)-\tfrac{1}{9}\times(-1)^3) \\ &= (\tfrac{2}{9})-(-\tfrac{2}{9}) = \tfrac{4}{9}\end{aligned}$$

8)

$$\frac{1}{x^2} = \frac{1}{x^3} \Rightarrow x = 1$$

$$\begin{aligned}\int_1^2 \frac{1}{x^2}\,dx - \int_1^2 \frac{1}{x^3}\,dx &= \int_1^2 (x^{-2} - x^{-3})\,dx \\ &= [\tfrac{x^{-1}}{-1} - \tfrac{x^{-2}}{-2}]_1^2 = [-\tfrac{1}{x} + \tfrac{1}{2x^2}]_1^2 \\ &= (-\tfrac{1}{2} + \tfrac{1}{2\times 2^2}) - (-\tfrac{1}{1} + \tfrac{1}{2\times 1^2}) \\ &= (-\tfrac{3}{8}) - (-\tfrac{1}{2}) = \tfrac{1}{8}\end{aligned}$$

9)

$$x(x-1)(x+1) = 2x(x-1)(x+1) \Rightarrow x = -1 \text{ or } x = 0 \text{ or } x = 1$$

$$\begin{aligned}\int_{-1}^0 2x(x-1)(x+1)\,dx - \int_{-1}^0 x(x-1)(x+1)\,dx& \\ = \int_{-1}^0 x(x-1)(x+1)\,dx &= \int_{-1}^0 x(x^2-1)\,dx \\ = \int_{-1}^0 (x^3 - x)\,dx &= [\tfrac{1}{4}x^4 - \tfrac{1}{2}x^2]_{-1}^0 \\ = (\tfrac{1}{4}\times 0^4 - \tfrac{1}{2}\times 0^2) - (\tfrac{1}{4}\times(-1)^4 - \tfrac{1}{2}\times(-1)^2)& \\ = (0) - (-\tfrac{1}{4}) = \tfrac{1}{4}&\end{aligned}$$

$$\begin{aligned}\int_0^1 2x(x-1)(x+1)\,dx - \int_0^1 x(x-1)(x+1)\,dx& \\ = \int_0^1 x(x-1)(x+1)\,dx &= \int_0^1 x(x^2-1)\,dx \\ = \int_0^1 (x^3 - x)\,dx &= [\tfrac{1}{4}x^4 - \tfrac{1}{2}x^2]_0^1 \\ = (\tfrac{1}{4}\times 1^4 - \tfrac{1}{2}\times 1^2) - (\tfrac{1}{4}\times 0^4 - \tfrac{1}{2}\times 0^2)& \\ = (-\tfrac{1}{4}) - (0) = -\tfrac{1}{4}&\end{aligned}$$

Therefore Area $= \frac{1}{4} - (-\frac{1}{4}) = \frac{1}{2}$

10)

$$\begin{aligned}&(x+1)(x-2)(x-3) = 3(x+1)(x-2)(x-4) \\ \Rightarrow\ &(x+1)(x-2)(x-3) - 3(x+1)(x-2)(x-4) = 0 \\ \Rightarrow\ &(x+1)(x-2)((x-3) - 3(x-4)) = 0 \\ \Rightarrow\ &(x+1)(x-2)(9-2x) = 0 \Rightarrow x = -1 \text{ or } x = 2 \text{ or } x = \tfrac{9}{2}\end{aligned}$$

$$\begin{aligned}\int_{-1}^2 3(x+1)(x-2)(x-4)\,dx - \int_{-1}^2 (x+1)(x-2)(x-3)\,dx& \\ = \int_{-1}^2 (3x^3 - 15x^2 + 6x + 24)\,dx&\end{aligned}$$

$$
\begin{aligned}
&\quad - \int_{-1}^{2} (x^3 - 4x^2 + x + 6)\, dx \\
&= \int_{-1}^{2} (2x^3 - 11x^2 + 5x + 18)\, dx \\
&= [\tfrac{1}{2}x^4 - \tfrac{11}{3}x^3 + \tfrac{5}{2}x^2 + 18x]_{-1}^{2} \\
&= (\tfrac{1}{2} \times 2^4 - \tfrac{11}{3} \times 2^3 + \tfrac{5}{2} \times 2^2 + 18 \times 2) \\
&\quad - (\tfrac{1}{2} \times (-1)^4 - \tfrac{11}{3} \times (-1)^3 + \tfrac{5}{2} \times (-1)^2 + 18 \times (-1)) \\
&= (\tfrac{74}{3}) - (-\tfrac{34}{3}) = 36
\end{aligned}
$$

$$
\begin{aligned}
\int_{2}^{9/2} (x+1)(x-2)(x-3)\, dx - \int_{2}^{9/2} 3(x+1)(x-2)(x-4)\, dx & \\
&= \int_{2}^{9/2} (x^3 - 4x^2 + x + 6)\, dx \\
&\quad - \int_{2}^{9/2} (3x^3 - 15x^2 + 6x + 24)\, dx \\
&= \int_{2}^{9/2} (-2x^3 + 11x^2 - 5x - 18)\, dx \\
&= [-\tfrac{1}{2}x^4 + \tfrac{11}{3}x^3 - \tfrac{5}{2}x^2 - 18x]_{2}^{9/2} \\
&= (-\tfrac{1}{2} \times (\tfrac{9}{2})^4 + \tfrac{11}{3} \times (\tfrac{9}{2})^3 - \tfrac{5}{2} \times (\tfrac{9}{2})^2 - 18 \times \tfrac{9}{2}) \\
&\quad - (-\tfrac{1}{2} \times 2^4 + \tfrac{11}{3} \times 2^3 - \tfrac{5}{2} \times 2^2 - 18 \times 2) \\
&= (-\tfrac{81}{32}) - (-\tfrac{74}{3}) = \tfrac{2125}{96}
\end{aligned}
$$

Therefore Area $= 36 + \frac{2125}{96} = \frac{5581}{96}$

11)

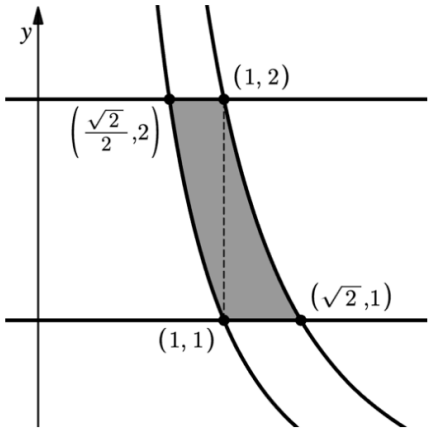
$$
\begin{aligned}
8x^3 + 6x^2 - 36x &= 81 - (8x^3 + 6x^2 - 36x) \\
\Rightarrow 2(8x^3 + 6x^2 - 36x) &= 81 \\
\Rightarrow 16x^3 + 12x^2 - 72x - 81 &= 0 \\
\Rightarrow (4x - 9)(2x + 3)^2 = 0 &\Rightarrow x = \tfrac{9}{4} \text{ or } x = -\tfrac{3}{2}
\end{aligned}
$$

$$
\begin{aligned}
\int_{-3/2}^{9/4} (81 - (8x^3 + 6x^2 - 36x))\, dx - \int_{-3/2}^{9/4} (8x^3 + 6x^2 - 36x)\, dx & \\
&= \int_{-3/2}^{9/4} (81 - 16x^3 - 12x^2 + 72x)\, dx \\
&= [81x - 4x^4 - 4x^3 + 36x^2]_{-3/2}^{9/4} \\
&= (81 \times \tfrac{9}{4} - 4 \times (\tfrac{9}{4})^4 - 4 \times (\tfrac{9}{4})^3 + 36 \times (\tfrac{9}{4})^2) \\
&\quad - (81 \times (-\tfrac{3}{2}) - 4 \times (-\tfrac{3}{2})^4 - 4 \times (-\tfrac{3}{2})^3 + 36 \times (-\tfrac{3}{2})^2) \\
&= (\tfrac{13851}{64}) - (-\tfrac{189}{4}) = \tfrac{16875}{64} = 263.671875
\end{aligned}
$$

12)

$$2 = \frac{1}{x^2} \Rightarrow x^2 = \tfrac{1}{2} \Rightarrow x = \pm\tfrac{\sqrt{2}}{2}$$

$$2 = \frac{2}{x^2} \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$
$$1 = \frac{1}{x^2} \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$
$$1 = \frac{2}{x^2} \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$$



$$\begin{aligned}(1 - \tfrac{\sqrt{2}}{2}) \times 2 - \int_{\sqrt{2}/2}^{1} \frac{1}{x^2}\, dx &= (2 - \sqrt{2}) - \int_{\sqrt{2}/2}^{1} x^{-2}\, dx \\ &= (2 - \sqrt{2}) - [-\tfrac{1}{x}]_{\sqrt{2}/2}^{1} \\ &= (2 - \sqrt{2}) - ((-\tfrac{1}{1}) - (-\tfrac{1}{\sqrt{2}/2})) \\ &= (2 - \sqrt{2}) - (-1 + \sqrt{2}) = 3 - 2\sqrt{2}\end{aligned}$$

$$\begin{aligned}\int_{1}^{\sqrt{2}} \frac{2}{x^2}\, dx - (\sqrt{2} - 1) \times 1 &= \int_{1}^{\sqrt{2}} 2x^{-2}\, dx - (\sqrt{2} - 1) \\ &= [-\tfrac{2}{x}]_{1}^{\sqrt{2}} - (\sqrt{2} - 1) \\ &= ((-\tfrac{2}{\sqrt{2}}) - (-\tfrac{2}{1})) - (\sqrt{2} - 1) \\ &= (2 - \sqrt{2}) - (\sqrt{2} - 1) = 3 - 2\sqrt{2}\end{aligned}$$

Therefore Area $= (3 - 2\sqrt{2}) + (3 - 2\sqrt{2}) = 6 - 4\sqrt{2}$